MEETING BRIEF · 25 SEPTEMBER 2026

Wildfire PINN pipeline report

Palisades Fire real-data build, quality-control findings, spread-rate baseline, and the evidence still needed before forecast claims.

The pipeline is now train-ready end to end: aligned environmental rasters, 14 filtered satellite-overpass perimeters, signed-distance targets, and a saved space-time spread-rate field. The remaining result is the real-data training run and its four held-out front-distance errors.

14 / 14
overpass perimeters built; none skipped

920
of 1,056 detections retained after spatial QC

48.1%
of grid cells classified as burnable

0.236
m/s day-1 median R, the highest reported day

READY TO REPORT

The real-data inputs are assembled

- 25 daily steps on a 178 × 290 grid at 90 m in EPSG:3310.
- Terrain, FBFM40 fuel, wind, and 100-hour fuel moisture are aligned.
- The first 10 overpasses are training targets; the last four remain held out.
- The initial condition and observation targets are signed-distance fields.

NOT YET A RESULT

Forecast skill is still unverified

- No completed real-data loss trace has been shown.
- No held-out front-distance values are available yet.
- The spread-rate field is a calibrated proxy, not full Rothermel.
- Coarse weather inputs limit spatial interpretation.

> **Recommended Friday headline:** “We have completed and sanity-checked the train-ready real-data pipeline. The next gate is quantitative: a reproducible training run and mean boundary-distance error on four unseen overpasses.”

Prepared from the current Palisades notebook, the filtered-perimeter output, the spread-rate diagnostics, and the project’s previously validated synthetic baselines. Results are labeled separately from estimates and pending work.

The pipeline now reaches a trainable physics model

The Palisades case spans 7–31 January 2025 on a 90 m California Albers grid. VIIRS observations cover 14 usable overpasses from 7–12 January; environmental forcing covers 25 daily steps.

VIIRS detections
confidence filter + overpass groups

→

Perimeter targets
outlier trim + alpha hull + SDF

→

Level-set PINN
physics + observations + holdout

LANDFIRE
elevation, slope, FBFM40

→

GridMET
wind speed/direction, fuel moisture

→

R(x,y,t)
first-order spread-rate proxy

The aligned stack has the variables needed for the current baseline

LAYER	SHAPE	ROLE IN MODEL
Elevation	178 × 290	Terrain context
Slope	178 × 290	Slope multiplier in R
FBFM40 fuel model	178 × 290	Fuel tier and moisture-of-extinction lookup
Wind speed / direction	25 × 178 × 290	Magnitude term in R; direction term in PDE
100-hour fuel moisture	25 × 178 × 290	Moisture damping in R

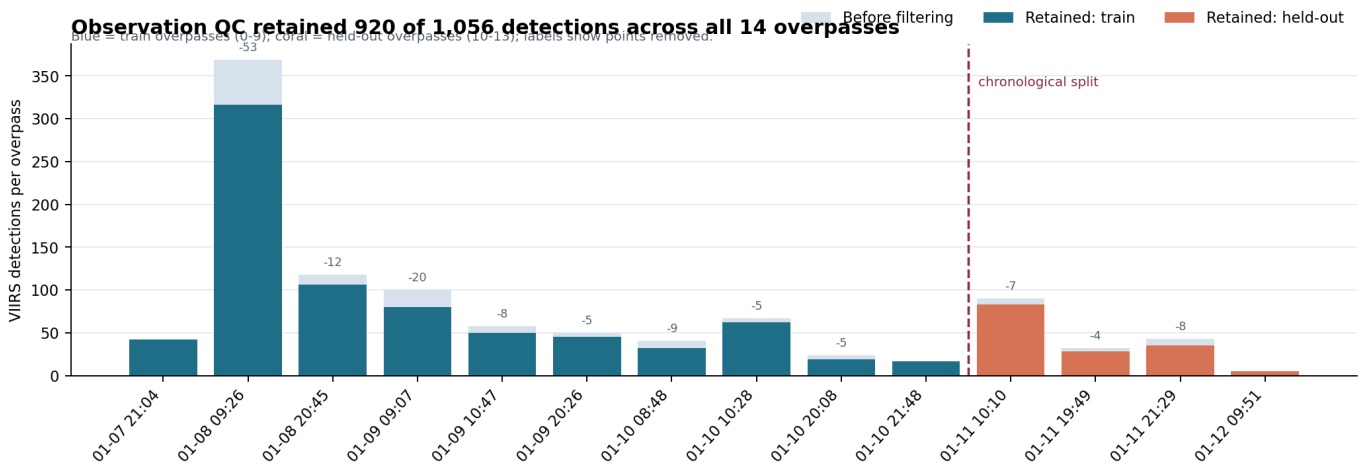


Figure 1. Spatial filtering removed 136 of 1,056 nominal/high-confidence detections (12.9%) while preserving every overpass. The largest trims occur during high-count Jan. 8 passes, where isolated detections most strongly inflated the hulls.

Observation construction

- Third-nearest-neighbor distance was computed in EPSG:3310.
- Points above median + 4 MAD were removed, never below four points.
- Alpha-hull perimeters were converted to signed-distance fields in meters.

Validation consequence

VIIRS hulls are instantaneous thermal-detection footprints, not cumulative burned-area perimeters. Their areas can shrink as ground cools. Boundary distance is therefore the defensible holdout metric; area error is not.

The spread-rate field passes a first sanity check, not a validation

The saved field **palisades_R.nc** converts fuel, wind speed, slope, and moisture into a daily scalar rate of spread. It supplies the physics residual at sampled space-time points.

```
R = C_REF × r_rel(fuel) × W(wind speed) × S(slope) × M(fuel moisture)
R_eff = max{0, R × [1 + C_WDIR (wind direction · front normal)]}
```

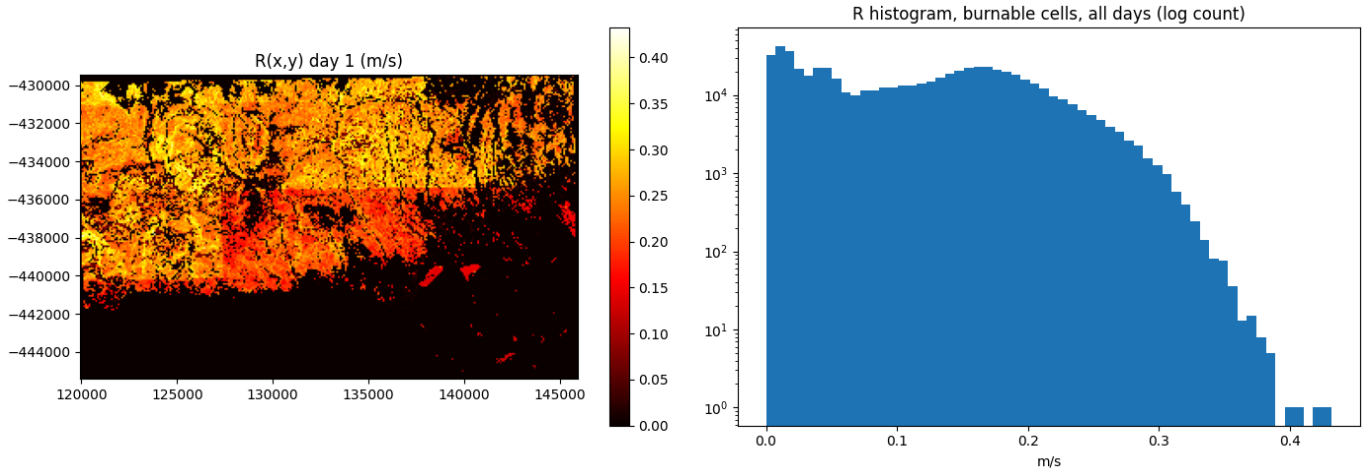


Figure 2. Day-1 spread-rate map and all-day distribution for burnable cells. The warm northern land region contrasts with the zero-rate ocean/non-burnable region. The block-like weather transition is a GridMET-resolution artifact rather than verified local detail.

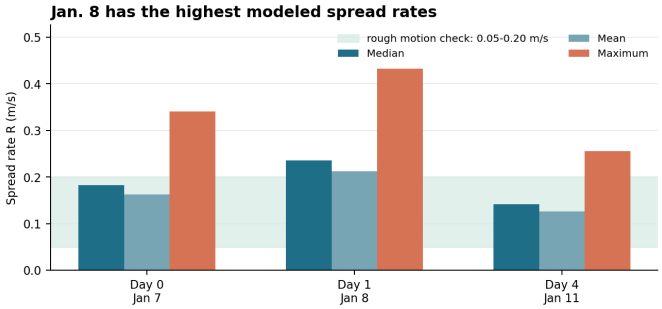


Figure 3. Jan. 8 is hottest in the reported sample, consistent with the observed blowup. The day-1 median is slightly above the rough motion band.

DAY	MEDIAN	MEAN	MAX
0 · Jan. 7	0.182	0.163	0.341
1 · Jan. 8	0.236	0.212	0.432
4 · Jan. 11	0.141	0.126	0.256

Units are m/s. Burnable fraction = 0.481. The rough observational check is approximately 0.05–0.20 m/s for kilometer-scale motion over roughly 12 hours.

POSITIVE SIGN

The high-spread day is Jan. 8, and the distribution is broad rather than constant.

CALIBRATION

C_REF = 0.177 m/s is the first knob to test; C_W, C_S, and C_WDIR also remain provisional.

SCOPE LIMIT

This is a heuristic first-order proxy. It must not be described as a full Rothermel implementation.

The model and evaluation protocol are fixed; the real run is pending

The corrected training cell uses a 4-layer, 128-unit tanh network with inputs (normalized time, x, y) and output normalized signed distance Φ. The earlier shape failure came from a stale **LevelSetPINN** instance rebinding the model name inside the real-training cell; the fixed notebook removes that collision and optimizes the standalone network.

```
Φ_τ + (R_eff T / L) ||∇Φ|| = 0
Loss = 10·L_IC + 1·L_PDE + 5·L_OBS
```

Training configuration

Architecture	3 → 128 × 4 → 1
Epochs	4,000
PDE points / epoch	8,192
IC points / epoch	4,096
OBS points / target	1,024 × 9
Optimizer	Adam, lr = 10^{-3}
Time horizon	4.53 days

Evaluation configuration

- Chronological training targets: overpasses 0–9.
- Unseen targets: overpasses 10–13.
- Extract the predicted Φ = 0 contour at each held-out time.
- Measure mean point-to-observed-boundary distance in meters.
- Report all four errors and their mean; report “no contour” as a failure, not zero.

The technical progression is real, but each stage answers a different question

STAGE	RESULT	INTERPRETATION
1D analytical baseline	0.23% relative L2	Training machinery and IC/BC formulation validated
1D weight ablation	1×: 0.13%; 10×: 0.23%; 100×: 0.47%	All under 0.5%; 100× over-constrains
2D synthetic circle	18.11% relative L2	Qualitatively correct expanding front; quantitative IC/representation error remains
2D Palisades	Training-ready	No forecast metric until the corrected cell finishes

> **CPU expectation:** approximately 20–30 minutes for the full 4,000-epoch run and held-out evaluation, based on a 2-CPU benchmark. This is an estimate, not an observed completion time.

Required output from the next run

Epoch loss lines at 500-step intervals; the four held-out errors for 11 Jan 10:10, 11 Jan 19:49, 11 Jan 21:29, and 12 Jan 09:51; mean held-out front error; saved checkpoint and the exact configuration.

Friday should lock the next experiment and the integration contract

> "The Palisades pipeline is complete through R(x,y,t), and observation QC preserved all 14 overpasses while removing 12.9% spatial outliers. The model is configured for a 10/4 chronological split. We are not calling it a forecast until the held-out boundary-distance errors are produced."

Main takeaways for the team

- **Data risk has dropped:** the raster stack, observation targets, and physics input now share one grid.
- **The validation metric is appropriate:** use front distance, because thermal hull area is not cumulative burned area.
- **The model is no longer blocked by plumbing:** the remaining gate is an executed training/holdout result.
- **The R field is replaceable:** its NetCDF interface can later accept full Rothermel or a learned/UNet-derived field without rewriting the PINN.

Main takeaways for the advisor

- **Method continuity:** the validated 1D machinery has been extended to a 2D level-set formulation.
- **Physics is explicit:** wind direction enters through the front normal, not through position angle.
- **Uncertainty is named:** heuristic fuel tiers, calibration constants, coarse weather, and VIIRS-derived hulls.
- **Success is measurable:** four unseen overpasses and one distance-based headline metric.

Decisions to request in the room

DECISION	WHY IT MATTERS NOW
Palisades 90 m vs. team CZU / 1 km contract	The prototype and shared system currently target different fires, extents, and resolutions.
Accept proxy R for baseline, or require full Rothermel next	This sets the order of calibration work versus forecast benchmarking.
Owner and schema for the production R field	The UNet–PINN handoff needs named channels, units, grid, time step, and nodata behavior.
Minimum acceptable holdout error	A threshold prevents post-hoc interpretation of the first real run.
Weather treatment	GridMET is ~4 km; domain-mean direction avoids pretending upsampled pixels are real detail.

Bottom line: Friday's defensible achievement is a complete, quality-controlled, train-ready real-data pipeline. A forecast claim begins only when the corrected run produces held-out boundary-distance errors.

The next work should convert readiness into evidence

The sequence below protects the held-out evaluation from tuning leakage and keeps each change attributable to one hypothesis.

GATE 1 · PRODUCE THE MISSING RESULT

Run the corrected real-training cell once

- Capture loss lines every 500 epochs for total, IC, PDE, and observation terms.
- Record the four held-out front-distance errors and their mean.
- Save the network checkpoint together with K_TRAIN, C_WDIR, loss weights, time horizon, and spatial scale.
- If no zero contour appears, record that as the result and debug before calibration.

GATE 2 · DIAGNOSE

Classify the first failure mode

- **Global offset:** SDF sign, normalization, or observation alignment.
- **Timing error:** time origin, overpass timestamps, or day indexing.
- **Speed error:** C_REF or directional gain.
- **Local shape error:** coarse weather, fuel tiers, or insufficient representation.

GATE 3 · TUNE DELIBERATELY

Ablate one quantity at a time

- C_REF first, because it scales the entire field.
- W_OBS next, because 5× has not been ablated.
- C_WDIR only after speed scale is plausible.
- Keep overpasses 10–13 untouched until the final comparison.

Artifact preservation checklist

ARTIFACT	PURPOSE	PERSISTENCE ACTION
palisades_stack.nc	Aligned environmental inputs	Copy out of the temporary runtime
palisades_perimeters.geojson	Filtered observation targets	Version with QC settings
palisades_R.nc	Current physics input	Store constants in metadata
palisades_sdf_train.npz	Training target cache	Keep with grid specification
levelset_pinn_real.pt	Model checkpoint	Save with run configuration
Loss + holdout outputs	Evidence for claims	Export text and figures

Stop condition before integration: one reproducible standalone forecast, at least one physics diagnostic, all four held-out errors, and a preserved configuration. Only then build the UNet-shaped raster fixture and compare coupling variants.

Appendix: observation audit and method notes

Per-overpass detection audit

OVERPASS	RAW	KEPT	REMOVED
01-07 21:04	42	42	0
01-08 09:26	369	316	53
01-08 20:45	118	106	12
01-09 09:07	100	80	20
01-09 10:47	58	50	8
01-09 20:26	50	45	5
01-10 08:48	41	32	9
01-10 10:28	67	62	5
01-10 20:08	24	19	5
01-10 21:48	17	17	0
01-11 10:10	90	83	7
01-11 19:49	32	28	4
01-11 21:29	43	35	8
01-12 09:51	5	5	0
Total	**1,056**	**920**	**136**

Key constants in the current baseline

C_REF	0.177 m/s
C_W	0.5
U_REF	2.237 m/s
C_S	1.0
C_WDIR	0.6
IC / PDE / OBS	10 / 1 / 5

Interpretive guardrails

- VIIRS alpha hulls are observation proxies, not authoritative incident perimeters.
- GridMET native resolution is about 4 km; the 90 m raster repeats coarse information.
- Wind direction is reduced to a per-day domain mean in the training cell.
- Near-zero burnable rates can arise from moisture damping; zero is also assigned to explicit non-burnable codes.
- Day-1 median R above 0.20 m/s is a calibration flag, not automatic evidence of error.

Data and method basis

- NASA FIRMS VIIRS 375 m active-fire detections
- LANDFIRE LF2020 elevation, slope, and FBFM40
- GridMET daily wind and 100-hour fuel moisture
- Scott & Burgan (2005) fuel-model moisture-of-extinction values
- RMRS-GTR-153 / NWCG S-290 spread-rate adjective bands
- Project notebook: WildfirePINN__5_fixed.ipynb

Current status timestamp: 23 September 2026. Training outputs are intentionally absent because no completed corrected real-data run had been reported at the time of writing.